MIL-Q-37779A
1 June 1978
SUPERSEDING
MIL-Q-37779 (DLA-DM)
5 July 1977

MILITARY SPECIFICATION

QUINIDINE POLYGALACTURONATE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Quinidine Polygalacturonate Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA).

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Quinidine Polygalacturonate Tablets, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type I.

S5.3 The following is added to this paragraph:

Shall be tablets containing 275 mg quinidine polygalacturonate equivalent to 166.25 mg of quinidine base within the designated assay limits for the tablets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

The tablets shall assay to contain not less than 94.0 percent (156.3 mg) and not more than 106.0 percent (176.2 mg) of the required amount of quinidine base when determined as specified in 4.3.1.

The quinidine polygalacturonate in the finished tablet shall comply with the identity test specified in 4.3.2.

S6.4.1 Appearance. Tablets may be slightly mottled.

S6.4.9.1 Disintegration. Tablets shall disintegrate in not more than 30 minutes, using USP method with simulated gastric fluid TS maintained at 37° ± 2° as the immersion fluid.

3.2 Ingredients.

3.2.1 Quinidine polygalacturonate. The quinidine polygalacturonate used in the manufacture of the tablets shall comply with the following:

3.2.1.1 Melting range. Shall melt with decomposition beginning at 180°C using USP procedure for class II material.

3.2.1.2 Ultra violet absorption spectrum. A UV spectrum in 0.1N hydrochloric acid shall exhibit a maxima at wave lengths of 318 and 348 nm similar to quinidine molecule at a concentration of 10 micrograms per ml.

3.2.1.3 Identification. The quinidine polygalacturonate used in the manufacture of the tablets shall be identified as specified in 4.5.3.1.

3.2.1.4 Unreacted quinidine. The quinidine polygalacturonate shall yield not more than 2.75 percent residue (quinidine) when determined as specified in 4.5.3.2.

3.2.1.5 Assay. The quinidine polygalacturonate shall assay to contain not less than 57 percent and not more than 63 percent quinidine base when determined as specified in 4.5.3.3.

3.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

* 3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated Items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests.

4.5.1 Assay for quinidine polygalacturonate in the finished tablet.

Weigh and finely powder no less than 20 tablets. Weigh accurately a portion of the powder equivalent to 100 mg of quinidine base and transfer into a 100 ml beaker, add 30 ml purified water and 3 ml diluted hydrochloric acid.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

Allow to stand with frequent agitation overnight. Transfer the mixture, filtering it through filter paper into a 200 ml volumetric flask and wash the beaker and filter paper with small portions of purified water, until the filtrate does not give a positive test for chloride ion. Make up to volume (200 ml) with 0.1N hydrochloric acid. Transfer a 2 ml aliquot (equivalent to 1.0 mg quinidine base) into a 100 ml volumetric flask and make up to volume with 0.1N hydrochloric acid. Prepare a quinidine base curve using quinidine Sulfate USP in 0.1N hydrochloric acid to make the following solutions: 4, 8, 10, 12 and 15 micrograms of quinidine per ml. Read the absorptions of these solutions at 348 nm in a suitable spectrophotometer using 0.1N hydrochloric acid as a blank. Read the absorption of the sample solution at 348 nm in a suitable spectrophotometer using 0.1N hydrochloric acid as a blank. Calculate the quantity of quinidine base in the sample from the prepared quinidine base curve.

4.5.2 Identification of quinidine polygalacturonate in the finished tablet.

Weigh a portion of the finely powdered tablets used in the assay for the tablets equivalent to 250 mg of quinidine polygalacturonate. Follow the procedure as described in 4.5.3.1 for the identification of quinidine and polygalacturonic acid.

4.5.3 Quinidine polygalacturonate (ingredient).

4.5.3.1 Identification.

To 250 mg of quinidine polygalacturonate add 3 ml of purified water, 1 ml normal sulfuric acid and 6 ml ethyl alcohol. Shake the mixture thoroughly and centrifuge. Decant the supernatant liquid, remove the solids and dry them in an oven at 105°C. The supernatant liquid shows a blue fluorescence with ultraviolet light. To 0.3 ml of the supernatant liquid diluted to 5 ml with purified water, add two drops of bromine TS and then 1 ml of ammonia TS. The liquid acquires an emerald green color due to the formation of thalleoquin; this is the identification for quinidine. Wash the dried precipitate from the original solution with alcohol and then add several milliliters of 1N sodium hydroxide and warm. The material dissolves in the sodium hydroxide solution with the development of a yellow color; this is the identification for polygalacturonic acid.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

4.5.3.2 Unreacted quinidine.

Add approximately 800 mg of quinidine polygalacturonate to a small beaker and add 5 ml of anhydrous chloroform. Mix the slurry well and pour the slurry onto a filter paper. Wash the beaker and filter paper with three 5 ml portions of anhydrous chloroform. Transfer all the chloroform washings to a tared beaker and evaporate to dryness. Dry to constant weight any residue at a temperature of 105°C. Weigh the residue and calculate the quantity if any.

4.5.3.3 Assay.

Dissolve a quantity of quinidine polygalacturonate equivalent to 100 mg of quinidine base in 0.1N hydrochloric acid and allow to stand overnight. Filter into a 100 ml volumetric flask and wash the precipitate with sufficient 0.1N hydrochloric acid to make exactly 100 ml of filtrate. Dilute 1 ml of filtrate to 100 ml with 0.1N hydrochloric acid. Read the absorption of this solution at 348 nm in a suitable spectrophotometer using 0.1N hydrochloric acid as a blank. Calculate the quantity of quinidine base from the previously calibrated quinidine base curve prepared from quinidine sulfate USP in 0.1N hydrochloric acid as directed in the assay for quinidine polygalacturonate in the tablets (see 4.5.1).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 500 tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twelve (12) units of issue shall be packed as specified in 5.4.4 Of PPP-C-186.

5.3.2 Level B. Twelve (12) units of issue shall be packed as specified in 5.4.3.2 Of PPP-C-186.

5.3.3 Level C. THe subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

* NOTES: (1) The National Stock Number need not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

* 5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-035-1720	QUINIDINE POLYGALACTURONATE TABLETS, 275 mg, 500s

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army-MD
Navy-MS
Air Force-03

Preparing activity:
DLA-DM

Project No. 6505-2020

Review activity:
DoD-MB

☆U.S. GOVERNMENT PRINTING OFFICE: 1978—703-122/3939

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-Q-37779A QUINIDINE POLYGALACTURONATE TABLETS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:29Z
Check the source to verify that this is the current version before use.